First-Principles Investigation of the Effect of Solutes on the Ideal Shear Resistance and Electronic Properties of Magnesium

P. Garg, I. Adlakha, and K. N. Solanki

Abstract

Solute addition is an effective way to enhance mechanical properties, especially in magnesium based alloys due to the limited number of slip systems available for deformation at the room temperature. Hence, the effects of various alloying elements on ideal shear resistance (ISR) across different slip systems of Mg were investigated using first-principles calculations. The addition of a Ce, Y, or Zr solute atom was found to decrease ISR, whereas the substitution of a Sn, Li, Al, or Zn atom increased the ISR of Mg. The most active slip system in Mg changed from the basal partial $(0001)[10\bar{1}0]$ to prismatic $(10\bar{1}0)[11\bar{2}0]$ upon substitution of a Ce, Y, or Zr solute atom, whereas the addition of Sn, Li, Al, or Zn solute atom had negligible effect on the plastic anisotropy. Furthermore, the electronic density of states and valence charge transfer provides a quantum insight into the underlying factors influencing the observed softening/strengthening behavior. For instance, the electronic density of states calculation shows that the contribution from *d* states of Ce, Y, and Zr solute atoms decreases the electronic structure stability of their respective solid solution, thereby enhancing slip activities. Theoretical analyses were also performed, and a shearability parameter was introduced to understand the implications of the observed variation in ideal shear resistance on the macroscopic behavior of Mg alloys.

Keywords

Magnesium • Ideal shear resistance • Electronic properties • First principles

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Introduction

The addition of alloying elements can be used to alter the mechanical properties of metals, especially magnesium (Mg) alloys which have been identified as a possible light weight alternative to iron- and aluminum-based alloys for structural applications [1–4]. The usage of Mg alloys at large scale has been restricted due to their limited room temperature plasticity, that stems from the insufficient activity of non-basal slip systems [5, 6]. Therefore, extensive studies have been performed to understand the effect of various solute additions [7–10], especially rare earth (RE) elements such as yttrium (Y), lanthanum (La) [11, 12], etc. on the room temperature strength and formability of Mg. A Mg–3 wt% Y alloy was reported to show enhanced room temperature ductility due to the activation of additional pyramidal <c + a> slip system [13]. However, atomistic simulations attributed this behavior to an increase in the critical resolved shear stress (CRSS) of the basal slip system due to the addition of Y leading to equal activation probabilities of basal and pyramidal slip systems [14]. On the contrary, experimental data showed strengthening of various slip systems along with extension twinning upon Y addition in Mg [15]. In Mg–Al alloys, enhanced ductility was reported due to grain boundary sliding [16] as well as a decrease in the basal plane stacking fault energy [17, 18]. The addition of Ce to Mg alloys was found to improve the room temperature ductility of Mg and the addition of Li and Zn solutes, respectively, increased the ductility of Mg alloys due to the activation of non-basal <c + a> and prismatic slip systems [19, 20]. Such divergent experimental and theoretical results reflect the ambiguous understanding of the effects of different solute atoms on the plastic deformation in Mg.

Theoretical studies, generally focused on calculating ideal strength (maximum attainable stress) and generalized stacking fault energy, have also been utilized to gain valuable insights into the effect of solute addition or various defects on the plastic deformation of a material [21–23]. The

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Table 1 Crystallographic orientations of the simulation cell adopted for the ideal shear resistance calculations

Slip plane	X	Y	Z
Basal	$11\bar{2}0$	$1\bar{1}00$	0001
Prismatic	$11\bar{2}0$	$1\bar{1}00$	0001
Pyramidal	$11\bar{2}0$	$1\bar{1}01$	Normal to x and y

strength of a real crystal is affected by various defects but the ideal shear strength/resistance for a defect-free crystal offers fundamental suggestions, especially for the dislocation nucleation or shearability of the crystal, i.e., minimum shear stress required to instigate plastic deformation in a pure crystal with no defects [24]. Therefore, the study of the ideal strength of a defect-free crystal can provide vital information to understand the mechanical properties as it dictates the stress required for dislocation nucleation in a pristine crystal [25]. The ideal shear strength (resistance) of many metals and ceramics was quantified by Ogata et al. [26] and was found to be related to the valance charge distribution and directional bonding between the atoms. Furthermore, quantifying directly the effect of various solutes on the ideal shear resistance in Mg solid solutions will provide novel insights into the bonding behavior of solute and host metal atoms and subsequent softening or hardening behavior.

In this study, first-principles calculations of the ideal shear resistance (τ) were performed for pure Mg and Mg–X solid solutions (X = Y, Al, Li, Zn, Zr, Sn, Ce) along different slip systems to study the effect of solute addition on shearability. The effect of different solutes on the ideal shear resistance of Mg can be can be inferred from the electronic structure calculations for different Mg solid solutions, thus providing critical information on alloy design or screening. The changes in electronic structure of Mg due to the presence of a solute atom were revealed with the help of electronic density of states (DOS) and valence charge transfer analysis. Furthermore, to understand the macroscopic implications of these findings, a shearability parameter was introduced to estimate the strengthening or softening effect of solutes on different slip systems of Mg.

Computational Technique

The ideal shear resistance of Mg alloys was examined using, the density functional theory (DFT)-based Vienna ab initio simulation package (VASP) [27]. The projector augmented wave (PAW) pseudopotentials [28] were used to represent Mg and various solute elements. The generalized gradient approximation (GGA) using the Perdew–Burke–Ernzerhof (PBE) [29] formulation was used to represent the exchange and correlation functions with an energy cutoff of 275 eV for Mg. Gamma pack k-point mesh of $16 \times 9 \times 10$ was selected after extensive k-point sampling convergence studies to perform Brillouin zone integrations. The ions were relaxed with a force and energy convergence criteria of 1 meV $Å^{-1}$ and 10^{-6} eV, respectively. The computed equilibrium structure parameters such as lattice constants, cohesive energy, and elastic constants were found to be in good agreement with the previously reported values [30, 31]. In the case of Mg–X solid solution, the supercell was obtained from 3 periodic lengths of a pure Mg unit cell along the different orientations given in Table 1.

Then, one desire solute atom was substituted on the shear plane in the supercell to maintain a solute concentration of 0.93 at.% (within the solubility limit for all solutes). The substituted solute atom could relax in all directions. Since the simple shear deformation of the simulation cell was carried out with periodic boundary conditions; the ideal shear resistance calculations are independent of the position of the solute atom in the simulation cell. The self-interactions between the substitutional atoms for the chosen supercell were negligible. For instance, the binding energy of Zr in Mg with 108 atoms (i.e., a $3 \times 3 \times 3$ supercell) and 256 atoms (i.e., a $4 \times 4 \times 4$ supercell) was found to be 0.01 eV/atom. Therefore, to minimize the computational requirements, we chose the $3 \times 3 \times 3$ supercell of Mg for this work. The cutoff energy and the k-point mesh density were selected based on the convergence studies for each solid solution supercell. The ideal shear resistance calculations were performed by applying simple shear strain along the slip direction in increments of 1% until yielding occurred. The atomic positions and basis vectors were fully relaxed at each step while keeping the applied strain fixed. Next, the electronic density of states for the unstrained configurations and valence charge density distribution was calculated to understand the variations in electronic structure of the solid solutions.

Results and Discussion

The addition of solute atoms can alter the plastic behavior of Mg by either reducing or increasing the plastic anisotropy based on the chemical and elastic interactions between the solute atom and the surrounding host lattice [32, 33]. To analyze the effects of various solute atoms (Y, Al, Li, Sn, Zn, Zr, and Ce) in Mg, the ideal shear resistance across different slip systems was quantified to understand the effect of solutes on the mechanical behavior (especially dislocation

nucleation) of Mg alloys. Furthermore, the suitability of the various solute atoms as effective means for providing solid solution strengthening/softening due to their electronic structure was thoroughly explored.

Ideal Shear Resistance

In the case of pure Mg, the ideal shear resistance along the basal partial $\{0001\}\langle 10\bar{1}0\rangle$, prismatic $\{10\bar{1}0\}\langle 11\bar{2}0\rangle$, pyramidal $\{10\bar{1}1\}\langle 11\bar{2}0\rangle$, and basal $\{0001\}\langle 11\bar{2}0\rangle$ slip systems was 1.48, 1.58, 1.62, and 1.76 GPa, respectively and was in agreement with previous findings [34] (Fig. 1). Furthermore, the basal partial slip system demonstrated least resistance to the shear deformation suggesting that it would be the most active slip system during plastic deformation of pure Mg. Note that throughout the paper ($\{0001\}\langle 11\bar{2}0\rangle$) slip system is referred as "basal" and ($\{0001\}\langle 10\bar{1}0\rangle$) slip system as "basal partial."

To understand the mechanical behavior of Mg alloys, the shear stress-strain response of various Mg–X solid solutions were compared to pure Mg along different slip systems. The addition of Ce, Y, or Zr in Mg was found to decrease the ideal shear resistance across different slip systems when compared to pure Mg (Fig. 1). For instance, the addition of Ce was found to decrease the ideal shear resistance of the basal partial slip system from 1.48 to 1.17 GPa; a similar decrease in ideal shear resistance, varying from 5 to 28%, was observed along the other slip systems in the Mg–Ce solid solution. Furthermore, the least resistant slip system changed from the basal partial to the prismatic slip system, due to the addition of Ce, Y, or Zr (Fig. 1). For instance, the addition of Ce was found to decrease the shear resistance of the basal partial slip system from 1.48 to 1.17 GPa; a similar decrease in shear resistance, varying from 5 to 28%, was observed along the other slip systems in the solid solution strengthened Mg alloy. Furthermore, the least resistant slip system changed from the basal partial to the prismatic slip system, due to the addition of Ce, Y or Zr (Fig. 1). In other words, addition of certain solutes (Ce, Y and Zr) in Mg can alter the macroscopic plastic behavior, thereby improving formability of Mg alloys.

Based on the promising findings of solute addition in Mg, the effect of other substitutional solute atoms, such as Sn, Li, Al, and Zn, was further explored. The addition of Sn led to a small increase in ideal shear resistance from 1.76 to 1.80 GPa along the basal slip system when compared to the pure Mg. However, a negligible effect on the ideal shear resistance was observed along the remaining slip systems (basal partial, prismatic and pyramidal) of Mg–Sn alloy as compared to pure Mg with the basal partial still as the least shear resistant slip system (Fig. 1). The addition of a Li atom resulted in a mixed effect on the ideal shear resistance along different slip systems of Mg–Li solid solution (Fig. 1), but the basal partial remained the least shear resistant slip system in the Mg–Li solid solution (Fig. 1h). Similarly, the solution strengthening of Mg by Al and Zn solute atoms increased the ideal shear resistance to 1.54 GPa and 1.74 GPa, respectively, when compared to pure Mg (1.48 GPa). Despite the addition of Sn, Li, Al, and Zn, basal partial was still the most dominant slip system in Mg alloys (Fig. 1).

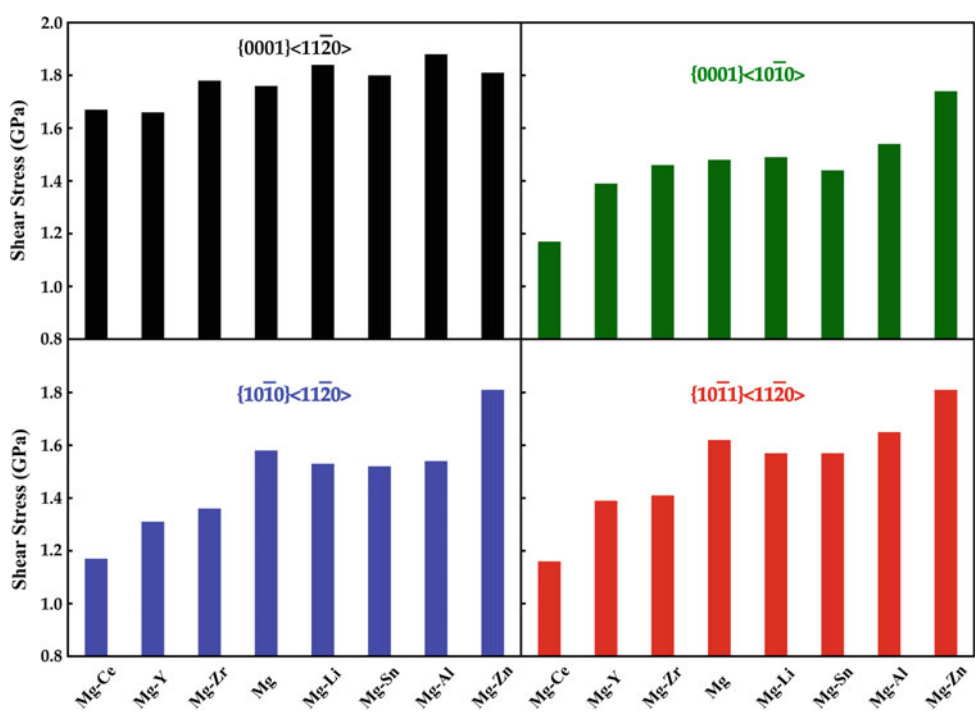

Fig. 1 The ideal shear resistance of pure Mg, Mg–Ce, Mg–Y, Mg–Zr, Mg–Li, Mg–Al, Mg–Sn, and Mg–Zn solid solution, respectively, along **a** basal $\{0001\}\langle 11\bar{2}0\rangle$, **b** basal partial $\{0001\}\langle 10\bar{1}0\rangle$, **c** prismatic $\{10\bar{1}0\}\langle 11\bar{2}0\rangle$ and **d** pyramidal $\{10\bar{1}1\}\langle 11\bar{2}0\rangle$ slip systems. *Note* the basal full dislocation is referred as basal and basal partial dislocation as basal partial

Density of States

The addition of a solute atom changes the surrounding atomic interactions, and thus, the examination of electronic structure can aid in understanding the strengthening/softening behavior in solution strengthened Mg alloys. Therefore, the effect of solute addition on the electronic density of states and valence charge transfer was examined to gain insight into the underlying factors influencing the ideal shear resistance of Mg–X solid solutions. The total electronic DOS for the various Mg solid solutions was obtained for the unstrained configuration (Fig. 2). The addition of Ce increased the DOS and decreased the electronic structure stability of the solid solution near the Fermi level due to the interaction of 5*d* states of Ce with 2*p* states of Mg. The DOS curves near the Fermi level for the solution strengthened Mg–Y and Mg–Zr alloys were found to have a noticeable shift due to the interaction of the partially filled 4*d* states of Y and Zr with 2*p* states of Mg, respectively. The *d* orbital contribution from Ce, Y and Zr solute atoms was found to cause an increase in the overall DOS for the Mg–X alloy at the Fermi level suggesting a decrease in the structural stability (i.e., shear resistance) of the solution strengthened alloys. On the other hand, addition of Sn, Li, Al, and Zn had negligible changes in the DOS of the solution strengthened alloy (Fig. 2b). Therefore, suggesting that the addition of Sn, Li, Al, and Zn solute atom had insignificant effect on the stability of the Mg–X alloy, respectively.

Valence Charge Transfer

The response of a material to externally applied load depends on the strength of bonds between the atoms. Thus,

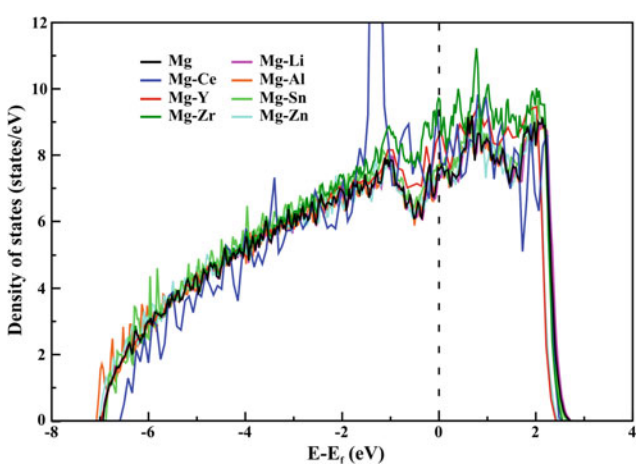

Fig. 2 Total density of states for pure Mg, Mg–Ce, Mg–Zr, Mg–Sn, Mg–Li, and Mg–Zn solid solutions, respectively, near the Fermi level (0 eV)

the valence charge transfer in various Mg solid solutions was quantified to provide insights into the redistribution of charge and strength of bonds. Furthermore, when the charge distribution is nonspherical/anisotropic the bonding between the atoms is directional leading to an increased ability to withstand shear deformation. Whereas the shear deformation occurs easily and the ideal shear resistance decreases when the charge density distribution is spherical/isotropic [35]. Therefore, it is crucial to understand the changes in valence charge distribution after the addition of solute atom and its effects on the ideal shear resistance of different slip systems. The valence charge density difference was calculated as:

$$\Delta\rho = \rho_{\mathrm{MgX}} - \rho_{\mathrm{Mg}} - \rho_{\mathrm{X}} \tag{1}$$

where $\Delta\rho$ represents the overall valence charge density difference in the solid solution while shearing, ρ_{MgX} is symbolic of the valence charge density of Mg–X solid solution, and ρ_{Mg} and ρ_{X} represent the valence charge density of non-interacting isolated Mg atoms and an X atom, respectively [36]. The solute addition had a localized effect in its surrounding region while the Mg–Mg bonding in the solid solution is unaffected, since neither charge redistribution nor charge transfer occurs in the bulk metal.

In Mg–Y solid solution, the valence electrons are attracted from the Y atom towards the more electronegative Mg atoms (Fig. 3a). Spherical charge density distribution suggests to the lowest ideal shear resistance on the prismatic slip system in Mg–Y (Fig. 3a3). In the case of the pyramidal slip system, a higher charge directionality was observed (Fig. 3a4) when compared to the prismatic slip system (Fig. 3a3), thereby indicating to the higher ideal shear resistance for the pyramidal slip system. The nonspherical and dense charge density distribution on the basal slip system in Mg–Y solid solution suggests to its higher shear resistance as compared to the basal partial slip system (Fig. 3a1–a2). The charge accumulated around the Zr atom is very small since Mg and Zr have similar electronegativity and atomic radius (Fig. 3b). The spherical charge density distribution on the prismatic slip system suggests to its lowest ideal shear resistance across various slip systems of Mg–Zr solid solution (Fig. 3b3). The increase in ideal shear resistance of Mg–Li solid solution (Fig. 3c) in comparison with Mg can be attributed to charge accumulation around the very small Li atom from the surrounding Mg atoms. The strengthening of bonds in the Mg–Al solid solution can be attributed to the valence *3p* electron of Al, which contributes to the charge density between the Al and surrounding Mg atoms. However, the isotropic charge density distribution around the Al atom on the prismatic slip system implies to a small decrease in its shear resistance (Fig. 3d3). Similarly, in Mg–Sn solid solution, the valence 5p electrons can contribute to the charge density between Sn and surrounding Mg

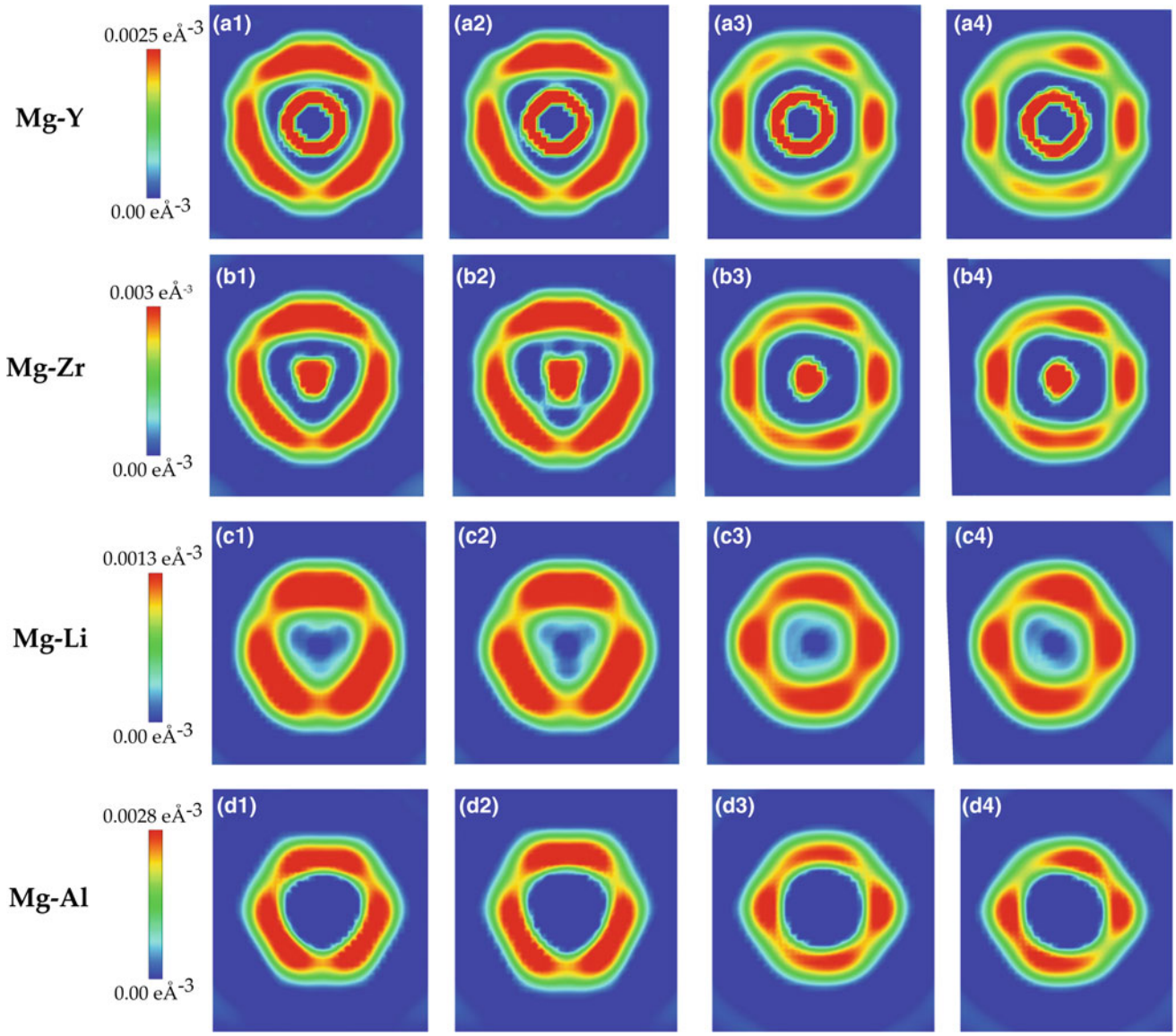

Fig. 3 The planar charge density ($eÅ^{-3}$) on the (1) basal, (2) basal partial, (3) prismatic, and (4) pyramidal slip system around the solute atom at the corresponding maximum shear stress in **a** Mg–Y, **b** Mg–Zr, **c** Mg–Li, and **e** Mg–Al solid solution. The solute atom is present in the center and surrounded by the Mg atoms at the corners of each square-shaped charge density contours

atoms. The anisotropic charge density distribution due to the directional spreading of *d* orbital electrons can contribute to the observed increase in the shear resistance of the Mg–Zn solid solution.

Shearability Parameter

Physically, the theoretical shear strength signifies the stress required to shear atomic planes in an ideal crystal. Frenkel [37] proposed a sinusoidal relationship between the shear stress and displacement to understand the microscopic plastic deformation process. Further, the criteria to predict the dominant slip system in a particular metal based on the stacking fault energy and elastic properties was first proposed by Legrand [38]. This parameter was further extended by Moitra et al. [39] to determine the effect of solute atoms in Mg with reference to pure Mg. However, the formability parameter in [39] is a function of the stacking fault energies and does not include the elastic properties. Thus, the proposed shearability parameter provides quantitative insights into the plastic behavior of Mg alloys by considering ideal shear resistance and elastic properties for different slip systems. The shearability parameter (R_τ) is defined as follows:

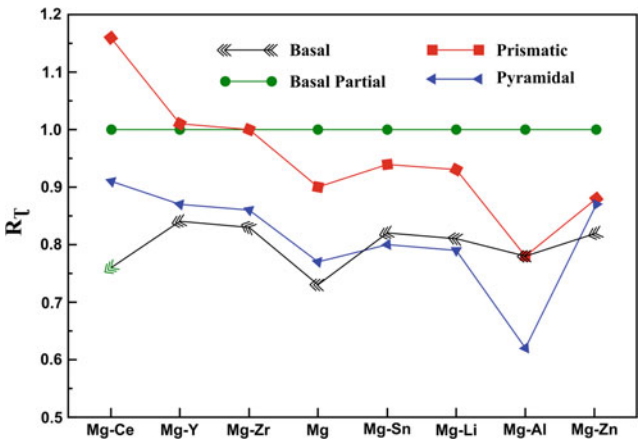

Fig. 4 Shearability parameter (R_τ) with respect to ideal shear resistance along various slip systems of Mg alloys

$$R_\tau = [\tau_{BP}/C_{BP}]/[\tau_P/C_P] \quad (2)$$

where, τ_{BP} and C_{BP} are the ideal shear resistance and the shear modulus along the basal partial slip system, respectively, and τ_P and C_P are the ideal shear resistance and shear modulus along the investigated slip system. The shearability parameter (R_τ) serves as an index of the solute strengthening/softening behavior for a slip system when compared to the primary slip system (i.e., basal partial for Mg). For instance, a R_τ value greater than or close to 1 indicates to a higher probability of activation for the non-basal slip systems. For example, in pure Mg, the R_τ value was 0.90 for the prismatic slip system, but the addition of Ce increased the R_τ value to 1.15. The addition of a Ce, Y, and Zr solute atom, was, respectively, found to increase the shearability parameter for non-basal slip systems (specifically the prismatic slip system) when compared to pure Mg (Fig. 4); thus, these solutes would improve the macroscopic formability of Mg alloys. On the other hand, the basal partial slip system was the most preferred with the addition of Sn, Li, Al, and Zn suggesting a negative impact on the formability of Mg alloys (Fig. 4). These observations from our theoretical calculations are found to be consistent with the experimentally and theoretically reported behavior, as observed in [13, 30, 39, 40].

Conclusions

A comprehensive first-principles study was carried out to understand the implications of solute addition on the plastic deformation of Mg alloys. Specifically, the effect of different solutes (Y, Al, Li, Zn, Zr, Sn, and Ce) on the ideal shear resistance along the various slip systems in Mg was quantified. Furthermore, the electronic structure (DOS) and nature of bonding (charge density contours) revealed the underpinning reasons for the observed shear response of various Mg alloys. The key findings of this work were:

1. The addition of Ce, Y, and Zr atoms decreased the ideal shear resistance along different slip systems of Mg alloys. Furthermore, the most active slip system changed from the basal partial to prismatic slip system leading to the observed softening up on addition of Ce, Y, and Zr solute atoms to Mg. On the other hand, the addition of Sn, Li, Al, and Zn atoms had no effect on the plastic anisotropy of Mg as the basal partial remained the most active slip system.
2. The DOS for the various Mg alloys revealed that the strong interactions of the *d* orbitals of Ce, Y, and Zr with *p* states of Mg contributed to a decrease in electronic structure stability and bond weakening that resulted in the reduction of ideal shear resistance across the various slip systems of Mg alloys.
3. The calculated values of R_τ show that shearability increased along the prismatic slip system of Mg–Ce, Mg–Y, and Mg–Zr solid solutions. On the other hand, the basal partial slip system had the maximum shearability for Mg–Sn, Mg–Li, Mg–Al, and Mg–Zn solid solutions resulting in no improvement in the plastic anisotropy.

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